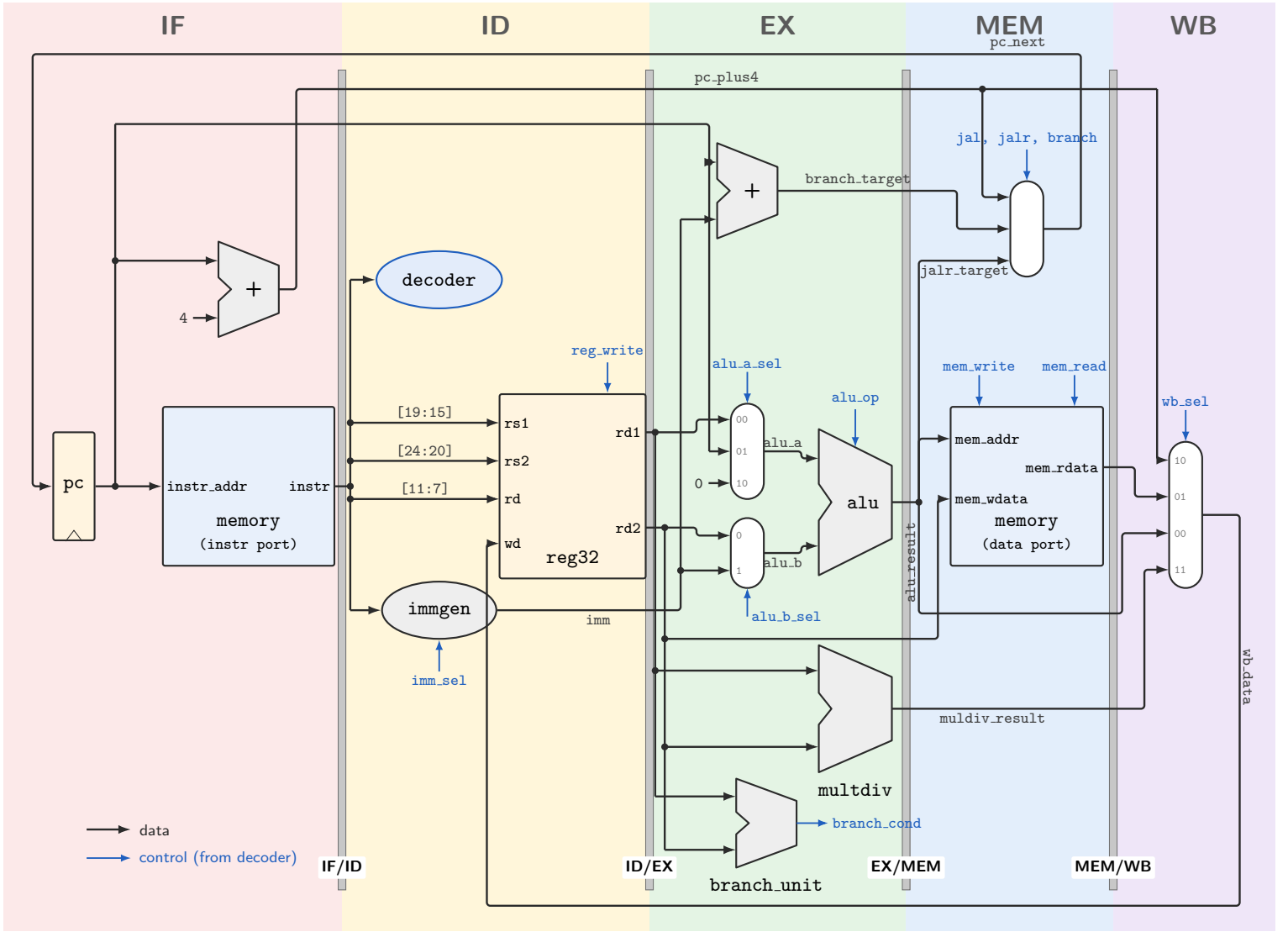
IF
ID
EX
MEM
WB
pc_next
pc_plus4
jal, jalr, branch
branch_target
jalr_target
decoder
4
reg_write
alu_a_sel
alu_op
mem_write
mem_read
wb_sel
[19:15]
[24:20]
[11:7]
rs1
rs2
rd
wd
rd1
rd2
reg32
pc
instr_addr
instr
memory
(instr port)
00
01
10
alu_a
0
alu
alu_b
0
1
alu_b_sel
alu_res
mem_addr
mem_rdata
mem_wdata
memory
(data port)
10
01
00
11
immgen
imm
imm_sel
muldiv_result
multdiv
branch_cond
branch_unit
wb_data
data
control (from decoder)
IF/ID
ID/EX
EX/MEM
MEM/WB